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DEVICE FOR DISPENSING VOLATILE SUBSTANCES

Abstract

A device for dispensing volatile substances to a surrounding environment includes a reservoir configured to hold a multi-component liquid mixture of non-volatile substances and volatile substances, a wick configured to be in selective communication with the liquid mixture in reservoir, an air inducing device that provides an airflow over the wick to release the volatile substances to a surrounding environment, and a rotating assembly coupled to the wick, the rotating assembly configured to rotate the wick to periodically replenish the wick with the liquid mixture prior to introducing the wick to the airflow.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to U.S. Provisional Patent Application No. 63/561,142, which was filed on Mar. 4, 2024 and entitled "Device for Dispensing Volatile Substances," the contents of which is hereby incorporated by reference in its entirety.

FIELD

[0002] The present disclosure relates to devices that dispense a volatile substance, including, but not limited to fragrances, insect repellent, or the like. More particularly, this disclosure is directed to a device that includes an assembly configured to rotate at least one wick between contact with a liquid mixture that includes the volatile substance and is absorbed by the wick, and an airflow generated by the device to disperse the volatile substance from the wick.

BACKGROUND

[0003] Dispersed volatile substances, like incense or insect repellent, is generally a liquid mixture that consists of a volatile liquid component mixed with a non-volatile (or less-volatile) liquid base. Current wick-based plug-in devices used for the spreading of volatile substances, like incense or insect repellent, in a local environment suffer from a decline in performance over time due to a reduction in the delivery of the volatile substances. This decrease in volatile substance distribution generally occurs because over time, the non-volatile (or less-volatile) components of the liquid mixture build up and block the wick, decreasing the effectiveness of the wick to disperse the desired volatile component of the liquid mixture.

[0004] The discussion above is merely provided for general background information and is not intended to be used as an aid in determining the scope of the claimed subject matter.

SUMMARY

[0005] In one aspect, the disclosure is directed to a device for dispensing volatile substances to a surrounding environment, the device including a reservoir configured to hold a multi-component liquid mixture of non-volatile substances and volatile substances; a wick configured to be in selective communication with the liquid mixture in reservoir; an air inducing device that provides an airflow over the wick to release the volatile substances to a surrounding environment; and a rotating assembly coupled to the wick, the rotating assembly configured to rotate the wick to periodically replenish the wick with the liquid mixture prior to introducing the wick to the airflow.

[0006] In another aspect, the disclosure is directed to a device for dispensing volatile substances to a surrounding environment, the device including a reservoir configured to hold a multi-component liquid mixture of non-volatile substances and volatile substances; a wick having a first portion and a second portion that are configured to be in selective communication with the mixture in reservoir; an air inducing device that provides an airflow over the first portion or the second portion of the wick to release the volatile substances to a surrounding environment; and a rotating assembly coupled to the wick, the rotating assembly configured to rotate so the wick cycles between a first position where the first portion of the wick is positioned to contact the mixture while the second portion of the wick is exposed to the airflow and a second position where the second portion of the wick is positioned to contact the liquid mixture while the first portion is exposed to the airflow.

[0007] In yet another aspect, the disclosure is directed to a device for dispensing volatile substances to a surrounding environment, the device including a reservoir configured to hold a multi-component liquid mixture of non-volatile substances and volatile substances; a wick having a first portion and a second portion that are configured to be in selective communication with the mixture in reservoir; and a rotating assembly coupled to the wick, the rotating assembly configured to rotate so the wick cycles between a first position where the first portion of the wick is positioned within the mixture while the second portion of the wick is exposed to the environment and a second position where the second portion of the wick is positioned within the liquid mixture while the first portion is exposed to the environment.

[0008] Other aspects of the invention will become apparent by consideration of the detailed description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. **1** is a schematic view of an embodiment of a device for dispensing volatile substances according to the present disclosure.

[0010] FIG. **2** is a perspective view of another example of an embodiment of a device for dispensing volatile substances according to the present disclosure.

[0011] FIG. **3** is a schematic side view of the device of FIG. **2**.

[0012] FIG. **4** is a table of measured data illustrating a rate of mass loss of a volatile compound provided in a liquid mixture according to five different case conditions, the data demonstrating the effectiveness of the device illustrated in FIG. **2** in dispensing volatile substances.

[0013] FIG. **5** is a graph illustrating the amount of volatile substances and non-volatile substances remaining in the device in each of the five case conditions shown in FIG. **4**.

[0014] FIG. **6** is a schematic side view of another example of an embodiment of a device for dispensing volatile substances according to the present disclosure.

[0015] FIG. **7** is a schematic top view of the device of FIG. **6**.

[0016] FIG. **8** is a perspective view of a wick for use with the device of FIG. **6**.

[0017] FIG. **9** is a schematic side view of another example of an embodiment of a device for dispensing volatile substances according to the present disclosure.

[0018] FIG. **10** is a schematic top view of the device of FIG. **9**.

[0019] FIG. **11** is a perspective view of a wick for use with the device of FIG. **9**.

DETAILED DESCRIPTION

[0020] Before any embodiments of the disclosure are explained in detail, it is to be understood that the invention is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The disclosure is capable of other embodiments and of being practiced or of being carried out in various ways.

[0021] The concepts disclosed in this discussion are described and illustrated by referring to exemplary embodiments. These concepts, however, are not limited in their application to the details of construction and the arrangement of components in the illustrative embodiments and are capable of being practiced or being carried out in various other ways. The terminology in this document is used for the purpose of description and should not be regarded as limiting. Words such as “including,” “comprising,” and “having” and variations thereof as used herein are meant to encompass the items listed thereafter, equivalents thereof, as well as additional items.

[0022] FIG. **1** schematically illustrates a device **10** for dispensing a volatile substance. Dispensing of a volatile substance during operation of the device **10** may deliver fragrances, incense, insect repellent, or other desirable volatile substances to a surrounding environment. It should be appreciated that the device **10** (or dispensing device **10**) that is schematically illustrated in FIG. **1** is an example of a device provided to illustrate one or more potential (or optional) components of the device. However, in some embodiments, the device **10** may include any combination of the components schematically illustrated in FIG. **1**.

[0023] With reference to FIG. **1**, the device **10** includes a reservoir **20** configured to hold a multi-component liquid mixture of at least a non-volatile substance and a volatile substance. It should be appreciated that the multi-component liquid mixture can includes a plurality of volatile substances and/or a plurality of non-volatile substances. One or more wicks **30** are configured to be in selective communication with the liquid mixture in reservoir **20**. The one or more wicks **30** are coupled to a rotation assembly **40**. The rotating assembly **40** is configured to move the wicks **30** relative to the reservoir **20**. More specifically, the rotating assembly **40** is configured to rotationally move the one or more wicks **30** relative to the assembly **40** into and out of selective contact with the liquid mixture in the reservoir **20**. An air inducing device **50** is in fluid communication with the one or more wicks **30**. The air inducing device **40** is configured to provide (or generate) an airflow. The airflow fluidly contacts the one or more wicks **30** after the at least one of the wicks **30** contacts the liquid mixture. Accordingly, the airflow generated by the air inducing device **40** is configured to release the volatile substance(s) from the one or more wicks **30** to the surrounding environment.

[0024] The device **10** can further include a controller **60** that is in communication with one or more components. In the illustrated embodiment, the controller **60** is in communication with the rotating assembly **40** and the air inducing device **50**. The controller **60** can also be in communication with a power supply **70**. The power supply **70** is configured to provide power to the rotating assembly **40**, the air inducing device **50**, and the controller **60**. In some embodiments, the device **10** may further include one or more sensors **80***a*, **80***b* (e.g., a reservoir sensor **80***a* and a wick sensor **80***b*) and a heating element **90** in communication with the reservoir **20**. The controller **60** can be in communication with the sensors **80***a*, **80***b*. The power supply **70** also provides power to the rotating assembly **40**, the controller **60**, and the heating element **90**. The power supply **70** may be an internal power source (e.g., a battery) or an external power source (e.g., a wall plug or the like). It should be appreciated that FIG. **1** illustrates communication connections by broken lines.

[0025] The reservoir **20** may be formed of one or more different materials, such as plastic, glass, or any other suitable material configured to hold the multi-component mixture. In some embodiments, the reservoir **20** may be fully enclosed (e.g., have a lid) or have an opening to the surrounding environment. The multi-component mixture in the reservoir **20** includes a combination of one or more non-volatile substances and one or more volatile substances or volatile organic compounds (VOCs). As a non-limiting example, the non-volatile substance(s) can include Hexadecane and Dodecane, while the volatile substance(s) can include Decane. However, it should be appreciated that any suitable or desired non-volatile substance(s) and/or volatile substance(s) can be combined to form the multi-component mixture. As described in the example provided in the detailed experimental results below, the multi-compound mixture utilizes liquid solvents that are close to alkane hydrocarbons in properties. During operation of the device **10**, the volatile substances are released from the liquid mixture and to the surrounding environment while the non-volatile substances remain in the liquid mixture. In some embodiments, diffuser oil formulations, insect repellent formulation, or the like may be developed with volatile substances for fragrances, insect repellent, or the like. The non-volatile substances may be odorless so they do not alter or obscure combined fragrances.

[0026] The one or more wicks **30** are formed of a porous material to facilitate the controlled absorption of the multi-component mixture in the reservoir and the controlled release of the volatile substances from the wicks **30**. In some embodiments, the wicks **30** may be formed of materials such as a ceramic, polymer, clay, carbon, wood, cellulose fiber, fiberglass, or the like. As described in the example provided in the detailed experimental results below, the wicks **30** are formed of a polymer because the multi-component liquid mixture used in the experiment has a comparatively lower surface energy compared to the polymers (e.g., PE, PP, PC). The construction of the wicks **30** ensures the wettability of pore surfaces of polymer wicks **30** by the multi-compound mixture, and thus leading to spontaneous imbibition of the multi-compound mixture into the wick **30**. However, it should be appreciated that the device **10** may be used with different multi-compound mixtures and materials of wicks to achieve desired absorption characteristics of the multi-compound mixture into the wicks and desired release of the volatile substances from the wicks. In the illustrated embodiment, the wicks **30** have a cylindrical or tubular shape.

[0027] The wicks **30** are coupled to the rotating assembly **40** such that the wicks are rotatable relative to the reservoir **20**. The rotating assembly **40** rotates the wicks **30** such that the wicks **30** are cycled between a first position and a second position. In the first position, the at least one of the wicks **30** is disposed within (or into contact with) the mixture in the reservoir **20**. In the second position, the at least one of the wicks **30** that was disposed within (or into contact with) the mixture in the first position is positioned within (or into contact with) the airflow induced by the air inducing device **50**. The airflow dispenses the volatile liquids from the at least one of the wicks **30** to the surrounding environment. After exposure to the airflow, the rotating assembly **40** returns the at least one of the wicks **30** from the second position back to the first position to replenish the wicks **30** with the mixture from within the reservoir **20**. This cycle then repeats to dispense the volatile substance from the mixture in the reservoir **20** into the surrounding environment. In addition, the rotating assembly **40** uses movement of the wicks **30** to stir the mixture in the reservoir **20**, further improving the efficiency of the coating of the mixture on an outer surface of the wicks **30**. When the wicks **30** are extracted from the reservoir **20**, the airflow from air inducing device **50** is blown over the wicks **30** to dispense the volatile substances into the surrounding environment. The remaining lesser volatile part of the mixture on the wicks **30** is mixed back into the liquid mixture in the reservoir when the wicks **30** are submerged back into the reservoir **20**. When the wicks **30** emerge from the reservoir **20** again, the wicks **30** are replenished with a new coating of the liquid mixture, which replenishes the volatile substance(s) on the wicks **30**. In particular, the rotation assembly **40** allows an outer surface of the wick(s) **30** to be coated with the mixture each time the wick(s) **30** enter the reservoir **20**, which reduces the amount of mixture that remains absorbed in the wick(s) **30**. In other words, the rotation mechanism **40** makes it so the wicks **30** do not rely on capillary action of the mixture to expose the volatile substances of the mixture to the airflow.

[0028] The controller **60** is in communication with the rotating assembly **40** and can adjust the rotational speed of the wicks **30**. In some embodiments, the controller **60** may continuously rotate the wicks **30** such that at least one wick is intermittently introduced to the airflow (in the second position) after the wick **30** is replenished with the mixture (in the first position). In some embodiments, the reservoir sensor **80***a* may detect the composition of the multi-compound mixture in the reservoir **20**. The controller **60** may adjust the speed of the rotating assembly **40** to release the optimal amount of volatile substances into the surrounding environment based on composition detected by the reservoir sensor **80***a*. In some embodiments, the controller **60** may intermittently slow or stop rotation of the wicks **30** so one of the wicks **30** is positioned in the airflow. In such an embodiment, the wick sensor **80***b* may monitor the moisture of the wick **30** exposed to the airflow. In response to the wick sensor **80***b* detecting the wick **30** reaching a predetermined moisture level, the controller **60** may instruct the rotating assembly **40** to rotate the wicks **30** (or increase the speed of rotation of the wicks **30**).

[0029] The air inducing device **50** provides an airflow over the wicks **30** to release the volatile substances to a surrounding environment. In some embodiments, the air inducing device **50** may be a fan coupled to the reservoir **20**. In other embodiments, the air inducing device **50** may be another component or system, such as an HVAC system, configured to generate airflow and that the device **10** is positioned within or adjacent to. As such, it should be appreciated that the air inducing device **50** may be integral with the device **10** or formed and powered separately from the device **10**.

[0030] The sensors **80***a*, **80***b* are in communication with the reservoir **20** to detect the composition of the multi-component mixture within the reservoir **20**. For example, the reservoir sensor **80***a* may detect the amount of volatile substance(s) remaining in the multi-component mixture. Once the amount of volatile substances reaches a predetermined amount, the reservoir sensor **80***b* may send a signal to the controller **60**. In response to the signal, the controller **60** may alert the operator of the device **10** that the multi-component mixture should be replaced with a fresh mixture (or additional volatile substance(s) should be added). In some embodiments, the wick sensor **80***b* may be in communication with the wicks **30** to detect one or more properties of the wicks **30**. As a non-limiting example, the wick sensor **80***b* can detect a moisture content of the wicks **30**, a weight (or change in weight) of the wicks **30**, or any other suitable property that can be used to detect a changing amount of multi-component mixture in the wicks **30**. In such embodiments, the controller **60** can responsively adjust the speed of the rotating assembly **40** in response to the properties of the wicks **30** detected by the wick sensor **80***b*. The heating element **90** is in communication with the reservoir **20** and is configured to heat the multi-component mixture. Raising the temperature of the multi-component mixture prior to coating the wick(s) **30** with the mixture can increases the rate that the volatile substances diffuse into the surrounding environment after being carried (or repositioned) into the airflow by the wick(s) **30**.

[0031] Now with reference to FIGS. **2** and **3**, a device **110** of an example embodiment is illustrated. The device **110** is similar to the device **10** illustrated in FIG. **1**. As such, like components are numbered with the same reference number plus "100". It should also be appreciated that all of the components associated with device **10** and shown in FIG. **1** may be used with the device **110** illustrated in FIG. **2**.

[0032] The device **110** includes a reservoir **120** configured to hold a multi-component liquid mixture **124** of at least a non-volatile substance and a volatile substance. A plurality of wicks **130** coupled to a support structure **128** (e.g., a shaft). A rotating assembly **140** (e.g., a stepper motor) is coupled to the reservoir **120**. An air inducing device **150** (e.g., a fan) coupled to the reservoir **120** and provides an airflow **126** (FIG. **3**) over one or more of the wicks **130** to release the volatile substances to a surrounding environment. A controller **160** is in communication with the rotating assembly **140** and the air inducing device **150**. The support structure **128** is coupled to the rotating assembly **140** such that the rotating assembly **140** rotates the support structure **128** relative to the reservoir **120**. The device **110** further includes wick holders **132** that are coupled to (or defined by) the support structure **128**. The wick holders **130** are each configured to slidably receive one of the wicks **130**.

[0033] In the illustrated embodiment, each wick holder **132** is supported on the support structure **128** and includes two receiving apertures that are respectively able to receive one of the plurality of wicks **130**. In some embodiments, the apertures of each wick holder **132** may be continuous such that a single wick is slidably received through the apertures. Each wick **130** responsively extends on opposing sides of the support structure **128**. In other embodiments, the wick holders **132** may be integrally formed with the support structure **128**. In other examples of embodiments, the wick holders **132** can be circumferentially offset relative to the support structure **128**. For example, each wick holder **132** can be radially offset from an adjacent wick holder **132** relative to the support structure **128**. Thus, each wick **130** can be radially offset from an adjacent wick **130** relative to the support structure **128**. Further, while the illustrated embodiment illustrates six total wicks **130***a-f*, it should be appreciated that the device **110** may include fewer wicks **130** (e.g., one, two, three, four, five) or more wicks **130** (e.g., seven, eight, etc.). In other examples of embodiments, the device **110** can include a plurality of wicks **130** or at least one wick **130**.

[0034] The wick holders **132** are arranged such that a first set of wicks **130***a-c* are arranged on a first side of the support structure **128** and a second set of wicks **130***d-f* are arranged on a second side of the support structure **128**. In other words, the first set of wicks **130***a-c* are positioned 180 degrees from the second set of wicks **130***d-f*. Stated another way, the first set of wicks **130***a-c* are generally parallel to the second set of wicks **130***d-f*. The orientation of the wicks **130***a-f* allows for the first set of wicks **130***a-c* to be exposed to the surrounding environment (and/or the associated airflow **126** generated by the air inducing device **150**) while the second set of wicks **130***d-f* are being replenished with the multi-component mixture (shown in FIG. **3**). It should be appreciated that in some embodiments the wicks **130***a*, **130***d* may be a single wick that extends through the support structure **128**. In such an embodiment, the first set of wicks **130***a-c* define a first portion of the wicks **130** and the second set of wicks **130***d-f* define a second portion of the wicks **130**.

[0035] Now with reference to FIG. **3**, the device **110** is illustrated in operation. During operation, the rotation assembly **140** is configured to rotate the support structure **128** in a first direction, shown by arrow **136**. The support structure **128** rotates relative to the reservoir **120** such that at least one wick **130***a* is exposed to the airflow **126** while at least one other wick **130***d* is submerged in the multi-component mixture **124** within the reservoir. The airflow **126** disperses the volatile substances on the wick **130***a* to the surrounding environment. As the support structure **128** continues to rotate, the wick **130***a* moves out of communication with the airflow **126** and into communication with the multi-component mixture **124** within the reservoir **120**. Concurrently, the at least one other wick **130***d* rotates out of communication with the multi-component mixture **124** within the reservoir **120** and into communication with the airflow **126**. This allows the wick **130***a* to replenish the liquid mixture on the wick **130***a*, while the other wick **130***d* is exposed to the airflow **126**. As described in detail above, in other examples of embodiments, the controller **160** may adjust the rotational speed of the support structure **128**. For example, the controller **160** may slow or intermittently stop rotation of the support structure **128** to effectively dispense the volatile substances from the wick(s) **130** positioned in the airflow **126** to the surrounding environment. Once the volatile substances are dispensed from the wick(s) **130**, such as by detection with one or more of the sensors **80***a*, **80***b* (shown in FIG. **1**), a predetermined amount of time, or other control step, the controller **160** may reinitiate rotation (or increase the speed of rotation) of the support structure **128**. In other examples of embodiment, the air inducing device **150** can be optional, as the volatile material may not require additional or directed airflow to facilitate dispersion into the environment.

[0036] FIGS. **4** and **5** illustrate experimental data relating to the device **110** during operation. In particular, the experimental data for the device **110**, which is provided in cases 4 and 5, illustrate the improved efficiency of dispersing volatile substances to the surrounding environment relative to other dispersing processes that do not include the device **110**. The experimental data includes five (5) separate test conditions, referred to as “cases,” where the percentage of volatile and non-volatile substances in the multi-component liquid mixture were tested after a constant time period for each case, here twenty-four (24) hours. In each experimental condition, Decane was used to represent the volatile substance. Dodecane and Hexadecane were used to represent the non-volatile substance. Each test condition started with the same formulation of the multi-component liquid mixture (Decane, Dodecane, and Hexadecane). The multi-component liquid mixture was then analyzed after the twenty-four (24) hour period to determine a quantity of volatile (Decane) and non-volatile (Dodecane and Hexadecane) remaining in the multi-component liquid mixture. Further, a mass loss during the time period was also determined relative to an initial composition of the mixture. In the experiment, the initial composition includes 49.428% volatile substances (measured as a percentage of Decane) and 50.542% non-volatile substances (measured as a percentage of 30.131% Dodecane and 20.411% Hexadecane; see FIG. **5**)

[0037] In case 1, the multi-component mixture is confined within a lidless reservoir in a calm, no-air-blown (or no air flow) environment. The mixture was introduced into the reservoir and was left undisturbed for a duration of 24 hours. When the liquid mixture is kept in an open reservoir, only the free surface area of the liquid is exposed to the surrounding air. As the mixture comes into direct contact with the air, the exposed surface area participates in evaporation, resulting in a steady loss of liquid mass over time. After 24 hours, the remaining mixture in the reservoir was analyzed to determine a reduction in mass and a relative ratio of remaining components, provided in a percentage. As shown in FIGS. **4** and **5**, the mixture experienced a mass loss of 0.034±0.011 gram/hour over the 24 hour period (see FIG. **4**) and the mixture had 46.836% volatile substances (measured as a percentage of Decane) remaining (see FIG. **5**).

[0038] In case 2, the multi-component mixture is confined within a lidless reservoir with three stationary wicks in the reservoir. This arrangement is similar to a commercially available “plug-in” type air freshener. When the wicks are immersed in the reservoir, the wicks absorb the mixture due to capillary action. As the wick absorbs the liquid, the wicks pulls the mixture from the reservoir and transports it down the length of the wick by capillary action, while also exposing some liquid to the outside air as it wets the cylindrical surface of the wicks. After 24 hours, the remaining mixture in the reservoir was analyzed to determine a reduction in mass and a relative ratio of remaining components, provided in a percentage. The recorded mass loss for this scenario is measured to be 0.064±0.011 gram/hour over the 24 hour period (see FIG. **4**), which is slightly higher than the mass loss in case 1. In addition, the mixture had 45.086% volatile substances remaining (measured as a percentage of Decane) (see FIG. **5**), which is less than the amount of volatile substances remaining in case 1.

[0039] In case 3, the multi-component mixture confined within a lidless reservoir with three rotating wicks in the reservoir, but no added airflow was introduced. The rotation of the wicks stirs the liquid multi-component mixture continuously thus maintaining a uniform concentration ratio in the reservoir. In addition, rotation of the wicks replaces the liquid mixture on the wicks with a fresh coat of the liquid mixture, thereby restoring the concentration of volatile substances on the wick. Unlike case 2, rotation of the wicks spreads the mixture out across a larger area of the wicks, which lead to an increased rate of volatile evaporation. The higher concentration of the volatile substances on the outer surface of the wicks increases the evaporation rate of the volatile substances. In addition, no additional airflow was introduced. Accordingly, evaporation was driven solely by exposure to the surrounding environment. The recorded mass loss for this scenario is measured to be 0.195±0.014 gram/hour over the 24 hour period (see FIG. **4**), which is three times higher than that in case 2. In addition, the mixture had 41.39% volatile substances remaining (measured as a percentage of Decane) (see FIG. **5**), which is less than the amount of volatile substances remaining in cases 1 and 2.

[0040] In case 4, the multi-component mixture confined within a lidless reservoir with three stationary wicks in the reservoir and a fan providing an airflow over the wicks was measured. With the stationary wicks projecting outward from the multi-component mixture, the fan is employed to generate airflow over a portion of the wicks, notably the portion of the wicks exposed above the reservoir. The airflow created by the fan serves to enhance the evaporation process by forcing air over moist wick surfaces and thus promoting evaporation. This continuous airflow ensures that the air around the wick remains virtually devoid of evaporated liquid vapor, allowing for improved evaporation efficiency of the volatile component. The recorded mass loss for this scenario is measured to be 0.397±0.091 gram/hour, demonstrating a higher mass loss compared to cases 1-3. In addition, the mixture had 5.29% volatile substances remaining (measured as a percentage of Decane) (see FIG. **5**), which is less than the amount of volatile substances remaining in cases 1-3.

[0041] In case 5, the multi-component mixture confined within a lidless reservoir with three rotating wicks in the reservoir and a fan providing an airflow over the wicks was measured (e.g., the illustrated device **110**). The experimental data showed a distinct difference between the mass-loss histories of the first 9 hours (i.e., 1.182±0.256 gm/hour) compared to the subsequent experimental time (i.e., 0.325±0.099 gm/hour). The observed difference in mass loss rates between the initial 9 hours and the subsequent period signifies a distinct change in the evaporation dynamics over time. The data illustrates that the evaporation process undergoes a transition or stabilization phase after the first 9 hours of the experiment. The two distinct trendlines indicate that the evaporation rate in the latter period, spanning the last 14 hours, is significantly lower compared to the initial phase. The recorded mass loss for this scenario during the first 9 hours is measured to be 1.182±0.256 gram/hour and 0.325±0.099 gram/hour for the remaining 14 hours. As such, the experimental data illustrates that the illustrated device **110** efficiently increases the dispersion of volatile substances and a higher mass loss over at least the first 9 hours of the experiment compared to cases 1-4. In addition, the mixture had 1.848% volatile substances remaining (measured as a percentage of Decane) (see FIG. **5**), which is less than the amount of volatile substances remaining in cases 1-4.

[0042] Now with reference to FIGS. **6-8**, another example of an embodiment of device **210** is illustrated. The device **210** is similar to the device **10**, **110** illustrated in FIGS. **1-3**. As such, like components are numbered with the same reference number plus "200". It should also be appreciated that all of the components associated with device **10** and device **110** and shown in FIGS. **1-3** may be used with the device **210** illustrated in FIGS. **6-8**.

[0043] The device **210** includes a reservoir **220** configured to hold a multi-component liquid mixture **224** of at least a non-volatile substance and a volatile substance. A wick **230** coupled to a support structure **228** (e.g., a shaft). A rotating assembly (not shown, e.g., a stepper motor) is coupled to the support structure **228**. An air inducing device **250** (e.g., a fan, an HVAC system, or the like) provides an airflow **226** (FIG. **6**) to a portion of the wick **230** to release the volatile substances to a surrounding environment. The support structure **228** is coupled to the rotating assembly such that the rotating assembly rotates the support structure **228** relative to the reservoir **220**.

[0044] As illustrated in FIG. **8**, the wick **230** includes an aperture **232** defined therein. The aperture **232** is configured to receive the support structure **228** to slidably support the wick **230**. The wick **230** has a disk-shaped geometry having a first side **235**, a second side **237**, and an outer circumference **239** extending between the first and second sides **235**, **237**. Further, a thickness T is defined between the first and second sides **235**, **237** of the wick **230**. In the illustrated embodiment, the thickness T is in a range of approximately 1 millimeter to approximately 10 millimeters, and more specifically in a range of approximately 1 millimeter to approximately 4 millimeters. In other embodiments. In other examples of embodiments, the thickness T can greater than 1 millimeter, greater than 10 millimeters, or can be any suitable or desired thickness or range of thicknesses. In some embodiments, the support structure **228** may include a plurality of wicks with a similar geometry as the wick **230**. Each of the plurality of wicks **230** can be spaced from an adjacent wick **230** along the support structure **228**. It should be appreciated that the disk-shaped construction increases a surface area of the wick **230** that is continuously exposed to the airflow **226**. This can advantageously increase the efficiency of the volatile substances dispersed from the wick **230** to the surrounding environment.

[0045] The wick **230** is arranged on the support structure **228** such that a first portion of the wick **230** is exposed to the surrounding environment (and/or the associated airflow **226** generated by the air inducing device **250**) while a second portion of the wick **230** is being replenished with the multi-component mixture (shown in FIG. **6**). Due to the continuous, disk-shaped construction of the wick **230**, the wick **230** is in continuous contact with the mixture **224** in the reservoir **220** to continuously coat the sides **235**, **237** and the outer circumference **239** of the wick **230** with the mixture.

[0046] With specific reference to FIG. **6**, the device **210** is illustrated in operation. During operation, the rotation assembly is configured to rotate the support structure **228** in a first direction, shown by arrow **236**. The support structure **228** rotates relative to the reservoir **220** such that the first portion of the wick **230** is exposed to the airflow **226** while the second portion of the wick **230** is submerged in the multi-component mixture **224** within the reservoir **220**. The airflow **226** disperses the volatile substances on the first portion of the wick **230** to the surrounding environment. As the support structure **228** continues to rotate, the first portion of the wick **230** moves out of communication with the airflow **226** and into communication with the multi-component mixture **224** within the reservoir **220**. Concurrently, the second portion of the wick **230** rotates out of communication with the multi-component mixture **224** within the reservoir **220** and into communication with the airflow **226**. This allows a portion of the wick **230** to continuously be replenished, while the remaining portion of wick **230** is exposed to the airflow **226**. As described in detail above, in other examples of embodiments, a controller may adjust the rotational speed of the support structure **228** to effectively dispense the volatile substances from the portion of the wick **230** positioned in the airflow **226** to the surrounding environment. Once the volatile substances are dispensed from the wick **230**, such as by detection with one or more of the sensors **80***a*, **80***b* (shown in FIG. **1**), by conclusion of a predetermined amount of time, or other suitable control step, the controller may reinitiate rotation (or change/increase the speed of rotation) of the support structure **228**. In other examples of embodiment, the air inducing device **250** can be optional, as the volatile material may not require additional or directed airflow to facilitate dispersion into the environment.

[0047] Now with reference to FIGS. **9-11**, another example of an embodiment of a device **310** is illustrated. The device **310** is similar to the device **10**, **110**, **210** illustrated in FIGS. **1-3** and FIGS. **6-8**. As such, like components are numbered with the same reference number plus “300”. It should also be appreciated that all of the components associated with device **10** and device **110** and shown in FIGS. **1-3** may be used with the device **310** illustrated in FIGS. **9-11**.

[0048] The device **310** includes a reservoir **320** configured to hold a multi-component liquid mixture **324** of at least a non-volatile substance and a volatile substance. A wick **330** is coupled to a support structure **328** (e.g., a shaft). A rotating assembly (not shown, e.g., a stepper motor) is coupled to the support structure **328**. An air inducing device **350** (e.g., a fan, HVAC system, or the like) provides an airflow **326** (FIG. **9**) to a portion of the wick **330** to release the volatile substances to a surrounding environment. The support structure **328** is coupled to the rotating assembly such that the rotating assembly rotates the support structure **328** relative to the reservoir **320**.

[0049] As illustrated in FIG. **11**, the wick **330** includes an aperture **332** defined therein. The aperture **332** is configured to receive the support structure **328** to slidably support the wick **330**. The wick **330** has a spherical, or generally ball-shaped, geometry defining an outer surface **339**. In the illustrated embodiment, the wick **330** is a sphere with a constant diameter. However, it should be appreciated that the wick **330** may be a spheroid, an oblong sphere, or other spherical shape having different diameters or radii. In some embodiments, the support structure **328** may include a plurality of wicks with a similar geometry as the wick **330** such that each wick **330** is spaced offset from an adjacent wick **330** along the support structure **328**.

[0050] The wick **330** is arranged on the support structure **328** such that a first portion of the wick **330** is exposed to the surrounding environment (and/or the associated airflow **326** generated by the air inducing device) while a second portion of the wick **330** is being replenished with the multi-component mixture (shown in FIG. **9**). Due to the continuous, spherical construction of the wick **330**, the wick **330** is in continuous contact with the mixture in the reservoir **320** to continuously coat the outer surface of the wick **330** with the mixture.

[0051] With specific reference to FIG. **9**, the device **310** is illustrated in operation. During operation, the rotation assembly is configured to rotate the support structure **328** in a first direction, shown by arrow **336**. The support structure **328** rotates relative to the reservoir **320** such that the first portion of the wick **330** is exposed to the airflow **326** while the second portion of the wick **330** is submerged in the multi-component mixture **324** within the reservoir **320**. The airflow **326** disperses the volatile substances on the first portion of the wick **330** to the surrounding environment. As the support structure **328** continues to rotate, the first portion of the wick **330** moves out of communication with the airflow **326** and into communication with the multi-component mixture **324** within the reservoir **320**. Concurrently, the second portion of the wick **330** rotates out of communication with the multi-component mixture **324** within the reservoir **320** and into communication with the airflow **326**. This allows a portion of the wick **330** to continuously be replenished, while the remaining portion of wick **330** is exposed to the airflow **326**. As described in detail above, in other examples of embodiments, a controller may adjust the rotational speed of the support structure **328** to effectively dispense the volatile substances from the portion of the wick **330** positioned in the airflow **326** to the surrounding environment. Once the volatile substances are dispensed from the wick **330**, such as by detection with one or more of the sensors **80***a*, **80***b* (shown in FIG. **1**), a predetermined amount of time, or other control step, the controller may reinitiate rotation (or change/increase the speed of rotation) of the support structure **328**. In other examples of embodiment, the air inducing device **350** can be optional, as the volatile material may not require additional or directed airflow to facilitate dispersion into the environment.

[0052] Although the invention has been described in detail with reference to certain preferred embodiments, variations and modifications exist within the scope and spirit of one or more independent aspects of the invention as described.

[0053] Various features of the invention are set forth in the following claims.

Claims

1. A device for dispensing volatile substances to a surrounding environment, the device comprising a reservoir configured to hold a multi-component liquid mixture of non-volatile substances and volatile substances; a wick configured to be in selective communication with the liquid mixture in reservoir; an air inducing device that provides an airflow over the wick to release the volatile substances to a surrounding environment; and a rotating assembly coupled to the wick, the rotating assembly configured to rotate the wick to periodically replenish the wick with the liquid mixture prior to introducing the wick to the airflow.

2. The device of claim 1, further comprising a support structure coupled to the rotating assembly, wherein the wick is coupled to the support structure, and wherein the rotating assembly rotates the support structure relative to the reservoir.

3. The device of claim 2, further comprising a wick holder coupled to the support structure, and wherein the wick holder is configured to slidably receive the wick.

4. The device of claim 1, further comprising a controller in communication with the rotating assembly and a power supply configured to provide power to the rotating assembly.

5. The device of claim 4, further comprising a sensor in communication with the reservoir, and wherein the sensor is configured to detect properties of the multi-component mixture within the reservoir.

6. The device of claim 5, wherein the sensor is configured to detect a composition of the multi-component liquid mixture within the reservoir, and wherein the controller is configured to alert an operator of the device that the multi-component mixture should be replaced with a fresh mixture when an amount of the volatile substances in the composition of the multi-component liquid mixture reaches a predetermined amount.

7. A device for dispensing volatile substances to a surrounding environment, the device comprising a reservoir configured to hold a multi-component liquid mixture of non-volatile substances and volatile substances; a wick having a first portion and a second portion that are configured to be in selective communication with the mixture in reservoir; an air inducing device that provides an airflow over the first portion or the second portion of the wick to release the volatile substances to a surrounding environment; and a rotating assembly coupled to the wick, the rotating assembly configured to rotate so the wick cycles between a first position where the first portion of the wick is positioned to contact the mixture while the second portion of the wick is exposed to the airflow and a second position where the second portion of the wick is positioned to contact the liquid mixture while the first portion is exposed to the airflow.

8. The device of claim 7, wherein the wick is formed of a single piece defining the first and second portions.

9. The device of claim 7, wherein the wick includes a first wick defining the first portion and a second wick defining the second portion.

10. The device of claim 7, further comprising a support structure coupled to the rotating assembly, wherein the wick is coupled to the support structure, and wherein the rotating assembly rotates the support structure relative to the reservoir.

11. The device of claim 10, further comprising a wick holder coupled to the support structure, and wherein the wick holder is configured to slidably receive the wick.

12. The device of claim 7, further comprising a controller in communication with the rotating assembly and a power supply configured to provide power to the rotating assembly.
13. The device of claim 12, further comprising a sensor in communication with the reservoir, and wherein the sensor is configured to detect properties of the multi-component mixture within the reservoir.
14. The device of claim 13, wherein the sensor is configured to detect a composition of the multi-component liquid mixture within the reservoir, and wherein the controller is configured to alert an operator of the device that the multi-component mixture should be replaced with a fresh mixture when an amount of the volatile substances in the composition of the multi-component liquid mixture reaches a predetermined amount.
15. A device for dispensing volatile substances to a surrounding environment, the device comprising a reservoir configured to hold a multi-component liquid mixture of non-volatile substances and volatile substances; a wick having a first portion and a second portion that are configured to be in selective communication with the mixture in reservoir; and a rotating assembly coupled to the wick, the rotating assembly configured to rotate so the wick cycles between a first position where the first portion of the wick is positioned within the mixture while the second portion of the wick is exposed to the environment and a second position where the second portion of the wick is positioned within the liquid mixture while the first portion is exposed to the environment.
16. The device of claim 15, further comprising a support structure coupled to the rotating assembly, wherein the wick is coupled to the support structure, and wherein the rotating assembly rotates the support structure relative to the reservoir.
17. The device of claim 16, further comprising a wick holder coupled to the support structure, and wherein the wick holder is configured to slidably receive the wick.
18. The device of claim 15, further comprising a controller in communication with the rotating assembly and a power supply configured to provide power to the rotating assembly.
19. The device of claim 15, wherein the wick is disc shaped, spherical in shape, or tubular in shape.
20. The device of claim 19, wherein the sensor is configured to detect a composition of the multi-component liquid mixture within the reservoir, and wherein the controller is configured to alert an operator of the device that the multi-component mixture should be replaced with a fresh mixture when an amount of the volatile substances in the composition of the multi-component liquid mixture reaches a predetermined amount.